```
=================================================================
                  START BATCH RUN: 06-01-2026
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----- Model 1 -----
----- Neural Network hyperparameters -----
Network configuration (neurons and activations) = {neurons:[38 64 32 16 8 4 1],
act_funcs:{relu, relu, relu, relu, relu, sigmoid}}
Parameter initialization method = he_normal
Positivity threshold = 0.885

----- Training hyperparameters -----
Learning rate = 0.0006
Momentum coefficient = 0.95
Number of epochs = 100
Minibatch size = 256
Regularization coefficient = 0.004

----- Results -----
Validation F2 = 0.6843
Training F2 = 0.6900

Validation Precision = 0.3899
Training Precision = 0.3923

Validation Recall = 0.8442
Training Recall = 0.8518



----- Model 2 -----
----- Neural Network hyperparameters -----
Network configuration (neurons and activations) = {neurons:[38 64 32 16 8 1],
act_funcs:{relu, relu, relu, relu, sigmoid}}
Parameter initialization method = he_normal
Positivity threshold = 0.875

----- Training hyperparameters -----
Learning rate = 0.0006
Momentum coefficient = 0.95
Number of epochs = 100
Minibatch size = 256
Regularization coefficient = 0.004

----- Results -----
Validation F2 = 0.6778
Training F2 = 0.6843

Validation Precision = 0.3834
Training Precision = 0.3870

Validation Recall = 0.8392
Training Recall = 0.8470



----- Model 3 -----
----- Neural Network hyperparameters -----
Network configuration (neurons and activations) = {neurons:[38 64 32 16 1],
act_funcs:{relu, relu, relu, sigmoid}}
Parameter initialization method = he_normal
Positivity threshold = 0.89

----- Training hyperparameters -----
Learning rate = 0.0006
Momentum coefficient = 0.95
Number of epochs = 100
Minibatch size = 256
Regularization coefficient = 0.004
```

```
----- Results -----
Validation F2 = 0.6755
Training F2 = 0.6796

Validation Precision = 0.3852
Training Precision = 0.3881

Validation Recall = 0.8324
Training Recall = 0.8368



----- Model 4 -----
----- Neural Network hyperparameters -----
Network configuration (neurons and activations) = {neurons:[38 64 32 1], act_funcs:{relu,
relu, sigmoid}}
Parameter initialization method = he_normal
Positivity threshold = 0.875

----- Training hyperparameters -----
Learning rate = 0.0006
Momentum coefficient = 0.95
Number of epochs = 100
Minibatch size = 256
Regularization coefficient = 0.004

----- Results -----
Validation F2 = 0.6714
Training F2 = 0.6754

Validation Precision = 0.3737
Training Precision = 0.3756

Validation Recall = 0.8386
Training Recall = 0.8438



----- Model 5 -----
----- Neural Network hyperparameters -----
Network configuration (neurons and activations) = {neurons:[38 64 1], act_funcs:{relu,
sigmoid}}
Parameter initialization method = he_normal
Positivity threshold = 0.855

----- Training hyperparameters -----
Learning rate = 0.0006
Momentum coefficient = 0.95
Number of epochs = 100
Minibatch size = 256
Regularization coefficient = 0.004

----- Results -----
Validation F2 = 0.6695
Training F2 = 0.6702

Validation Precision = 0.3634
Training Precision = 0.3634

Validation Recall = 0.8482
Training Recall = 0.8495



----- Model 6 -----
----- Neural Network hyperparameters -----
Network configuration (neurons and activations) = {neurons:[38 128 64 32 16 1],
act_funcs:{relu, relu, relu, relu, sigmoid}}
```

Parameter initialization method = he_normal
Positivity threshold = 0.89

----- Training hyperparameters -----
Learning rate = 0.0006
Momentum coefficient = 0.95
Number of epochs = 100
Minibatch size = 256
Regularization coefficient = 0.004

----- Results -----
Validation F2 = 0.6830
Training F2 = 0.6907

Validation Precision = 0.3884
Training Precision = 0.3933

Validation Recall = 0.8431
Training Recall = 0.8521

----- Model 7 -----
----- Neural Network hyperparameters -----
Network configuration (neurons and activations) = {neurons:[38 128 64 32 1], act_funcs:{relu, relu, relu, sigmoid}}
Parameter initialization method = he_normal
Positivity threshold = 0.875

----- Training hyperparameters -----
Learning rate = 0.0006
Momentum coefficient = 0.95
Number of epochs = 100
Minibatch size = 256
Regularization coefficient = 0.004

----- Results -----
Validation F2 = 0.6810
Training F2 = 0.6861

Validation Precision = 0.3781
Training Precision = 0.3800

Validation Recall = 0.8519
Training Recall = 0.8591

----- Model 8 -----
----- Neural Network hyperparameters -----
Network configuration (neurons and activations) = {neurons:[38 256 128 1], act_funcs:{relu, relu, sigmoid}}
Parameter initialization method = he_normal
Positivity threshold = 0.89

----- Training hyperparameters -----
Learning rate = 0.0006
Momentum coefficient = 0.95
Number of epochs = 100
Minibatch size = 256
Regularization coefficient = 0.004

----- Results -----
Validation F2 = 0.6797
Training F2 = 0.6852

Validation Precision = 0.3978
Training Precision = 0.4011

```
Validation Recall = 0.8262
Training Recall = 0.8327



----- Model 9 -----
----- Neural Network hyperparameters -----
Network configuration (neurons and activations) = {neurons:[38 256 1], act_funcs:{relu,
sigmoid}}
Parameter initialization method = he_normal
Positivity threshold = 0.875

----- Training hyperparameters -----
Learning rate = 0.0006
Momentum coefficient = 0.95
Number of epochs = 100
Minibatch size = 256
Regularization coefficient = 0.004

----- Results -----
Validation F2 = 0.6728
Training F2 = 0.6752

Validation Precision = 0.3907
Training Precision = 0.3923

Validation Recall = 0.8211
Training Recall = 0.8237





=================================================================
                  START BATCH RUN: 08-01-2026
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----- Model 1 -----
----- Neural Network hyperparameters -----
Network configuration (neurons and activations) = {neurons:[38 64 32 16 8 4 1],
act_funcs:{relu, relu, relu, relu, relu, sigmoid}}
Parameter initialization method = he_normal
Positivity threshold = 0.87

----- Training hyperparameters -----
Learning rate = 0.001
Momentum coefficient = 0.95
Number of epochs = 100
Minibatch size = 256
Regularization coefficient = 0.004

----- Results -----
Validation F2 = 0.6879
Training F2 = 0.6974

Validation Precision = 0.3859
Training Precision = 0.3897

Validation Recall = 0.8569
Training Recall = 0.8698



----- Model 2 -----
----- Neural Network hyperparameters -----
Network configuration (neurons and activations) = {neurons:[38 128 64 32 16 1],
act_funcs:{relu, relu, relu, relu, sigmoid}}
Parameter initialization method = he_normal
Positivity threshold = 0.88
```

```
----- Training hyperparameters -----
Learning rate = 0.001
Momentum coefficient = 0.95
Number of epochs = 100
Minibatch size = 256
Regularization coefficient = 0.004

----- Results -----
Validation F2 = 0.6900
Training F2 = 0.6993

Validation Precision = 0.3863
Training Precision = 0.3901

Validation Recall = 0.8592
Training Recall = 0.8721



----- Model 3 -----
----- Neural Network hyperparameters -----
Network configuration (neurons and activations) = {neurons:[38 128 64 32 1],
act_funcs:{relu, relu, relu, sigmoid}}
Parameter initialization method = he_normal
Positivity threshold = 0.875

----- Training hyperparameters -----
Learning rate = 0.001
Momentum coefficient = 0.95
Number of epochs = 100
Minibatch size = 256
Regularization coefficient = 0.004

----- Results -----
Validation F2 = 0.6876
Training F2 = 0.6935

Validation Precision = 0.3824
Training Precision = 0.3860

Validation Recall = 0.8592
Training Recall = 0.8660



=================================================================
                  START BATCH RUN: 08-01-2026
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----- Model 1 -----
----- Neural Network hyperparameters -----
Network configuration (neurons and activations) = {neurons:[38 128 64 32 16 1],
act_funcs:{relu, relu, relu, relu, sigmoid}}
Parameter initialization method = he_normal
Positivity threshold = 0.875

----- Training hyperparameters -----
Learning rate = 0.002
Momentum coefficient = 0.95
Number of epochs = 100
Minibatch size = 256
Regularization coefficient = 0.004

----- Results -----
Validation F2 = 0.6930
Training F2 = 0.7029

Validation Precision = 0.3955
```

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Training Precision = 0.4006

Validation Recall = 0.8547
Training Recall = 0.8677



----- Model 2 -----
----- Neural Network hyperparameters -----
Network configuration (neurons and activations) = {neurons:[38 128 64 32 16 1],
act_funcs:{relu, relu, relu, relu, sigmoid}}
Parameter initialization method = he_normal
Positivity threshold = 0.9

----- Training hyperparameters -----
Learning rate = 0.002
Momentum coefficient = 0.95
Number of epochs = 100
Minibatch size = 128
Regularization coefficient = 0.004

----- Results -----
Validation F2 = 0.6963
Training F2 = 0.7027

Validation Precision = 0.4010
Training Precision = 0.4045

Validation Recall = 0.8547
Training Recall = 0.8622



=================================================================
                  START BATCH RUN: 09-01-2026
=================================================================

----- Model 1 -----
----- Neural Network hyperparameters -----
Network configuration (neurons and activations) = {neurons:[38 128 64 32 16 1],
act_funcs:{relu, relu, relu, relu, sigmoid}}
Parameter initialization method = he_normal
Positivity threshold = 0.88

----- Training hyperparameters -----
Learning rate = 0.003
Momentum coefficient = 0.95
Number of epochs = 30
Minibatch size = 256
Regularization coefficient = 0.004

----- Results -----
Validation F2 = 0.6845
Training F2 = 0.6931

Validation Precision = 0.3784
Training Precision = 0.3821

Validation Recall = 0.8620
Training Recall = 0.8742



----- Model 2 -----
----- Neural Network hyperparameters -----
Network configuration (neurons and activations) = {neurons:[38 128 64 32 16 1],
act_funcs:{relu, relu, relu, relu, sigmoid}}
Parameter initialization method = he_normal
Positivity threshold = 0.895
```

```
----- Training hyperparameters -----
Learning rate = 0.002
Momentum coefficient = 0.95
Number of epochs = 30
Minibatch size = 256
Regularization coefficient = 0.004

----- Results -----
Validation F2 = 0.6831
Training F2 = 0.6902

Validation Precision = 0.3807
Training Precision = 0.3840

Validation Recall = 0.8527
Training Recall = 0.8624




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                  START BATCH RUN: 09-01-2026
=================================================================

----- Model 1 -----
----- Neural Network hyperparameters -----
Network configuration (neurons and activations) = {neurons:[38 128 64 32 16 1],
act_funcs:{relu, relu, relu, relu, sigmoid}}
Parameter initialization method = he_normal
Positivity threshold = 0.89

----- Training hyperparameters -----
Learning rate = 0.003
Momentum coefficient = 0.95
Number of epochs = 50
Minibatch size = 256
Regularization coefficient = 0.004

----- Results -----
Validation F2 = 0.6896
Training F2 = 0.6995

Validation Precision = 0.3987
Training Precision = 0.4044

Validation Recall = 0.8451
Training Recall = 0.8569



----- Model 2 -----
----- Neural Network hyperparameters -----
Network configuration (neurons and activations) = {neurons:[38 128 64 32 16 1],
act_funcs:{relu, relu, relu, relu, sigmoid}}
Parameter initialization method = he_normal
Positivity threshold = 0.885

----- Training hyperparameters -----
Learning rate = 0.002
Momentum coefficient = 0.95
Number of epochs = 50
Minibatch size = 256
Regularization coefficient = 0.004

----- Results -----
Validation F2 = 0.6883
Training F2 = 0.6965
```

Validation Precision = 0.3896
Training Precision = 0.3918

Validation Recall = 0.8527
Training Recall = 0.8654

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START BATCH RUN: 10-01-2026
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----- Model 1 -----
----- Neural Network hyperparameters -----
Network configuration (neurons and activations) = {neurons:[38 128 64 32 16 1], act_funcs:{relu, relu, relu, relu, sigmoid}}
Parameter initialization method = he_normal
Positivity threshold = 0.89

----- Training hyperparameters -----
Learning rate = 0.003
Momentum coefficient = 0.95
Number of epochs = 70
Minibatch size = 256
Regularization coefficient = 0.004

----- Results -----
Validation F2 = 0.6925
Training F2 = 0.7021

Validation Precision = 0.3945
Training Precision = 0.3985

Validation Recall = 0.8541
Training Recall = 0.8684

----- Model 2 -----
----- Neural Network hyperparameters -----
Network configuration (neurons and activations) = {neurons:[38 128 64 32 16 1], act_funcs:{relu, relu, relu, relu, sigmoid}}
Parameter initialization method = he_normal
Positivity threshold = 0.895

----- Training hyperparameters -----
Learning rate = 0.002
Momentum coefficient = 0.95
Number of epochs = 70
Minibatch size = 256
Regularization coefficient = 0.004

----- Results -----
Validation F2 = 0.6894
Training F2 = 0.6991

Validation Precision = 0.4074
Training Precision = 0.4111

Validation Recall = 0.8361
Training Recall = 0.8494